Ashirathee Inya

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Hands-on experience building software, data, and AI systems in the financial services sector, from requirements gathering and design to testing and deployment. Delivered a FastAPI and PostgreSQL service that a business team runs in production each month, built Python pipelines that collect and clean irregular data, and developed LLM applications with retrieval, agents, and evaluation against real user questions. Works across Python, SQL, REST APIs, Docker, and GitLab CI/CD, keeps sensitive data out of LLM calls under PDPA, and presents technical results to engineers and executives.

Education

Silpakorn University, Sanam Chandra Palace Campus *Jun. 2022 – Apr. 2026*

Bachelor of Engineering in Electronics and Computer System Engineering

Relevant Coursework: Artificial Intelligence, Database Design, Data Structures & Algorithms, Statistics, Pattern Recognition, Embedded Systems

Experience

Finansia Syrus Securities Public Company Limited

AI Engineer (Part-time) *Oct. 2025 – Aug. 2026*

- **Delivered and deployed a fee-reconciliation service** (FastAPI, PostgreSQL) that turned the Product team's manual monthly comparison into a production upload-and-export flow, producing a three-sheet Excel report from an uploaded trade file and an API-sourced staff roster.
- **Built and documented REST APIs (FastAPI, OpenAPI)** agreed with the team's frontend developer, with pytest and Postman checks.
- **Designed the fee service's PostgreSQL schema, ER diagram, and migrations** with senior engineer review and validation against real data, and set up a Docker-based GitLab CI/CD pipeline that carried each change through local, dev, and UAT testing before the production release.
- **Parsed irregular trade sheets** where one record's figures spanned several sub-rows, and matched names with 95%+ fuzzy similarity to absorb typos, with Decimal-precision calculations and audit-traceable storage.
- **Developed regulation-knowledge agents** in the team's internal multi-agent Generative AI service, where agents work alone or pass tasks to one another, letting finance staff query regulator documents in chat, and tuned each agent's scope, prompts, and model settings.
- **Built scheduled Python scraping pipelines** (Selenium, BeautifulSoup) that fed the agents' knowledge base with new regulator documents.
- **Benchmarked AI tools through proofs of concept**, testing open-source answer engines, search APIs, and the built-in web search of major LLMs against a paid AI search service on answer quality and cost per case.

AI Engineer Intern *May 2025 – Oct. 2025*

- **Initiated and built a customer support chatbot MVP** for financial platform and app inquiries, owning it end-to-end from requirements, architecture, and tool selection to testing and deployment, and presented it to the team lead, CTO, and academic advisor.
- **Built a document-processing pipeline** for 300+ pages of public Thai-language manuals and scraped company web pages, where key instructions sat in tables, infographics, and screenshots, converting each section into LLM-readable Markdown with Gemini image descriptions.
- **Chose the stack from hands-on tests** of PDF parsers, Thai OCR (Tesseract), embedding models (BGE-M3), and vector databases (Qdrant).

Projects

Customer Support Chatbot for a Financial Trading Platform

- **Designed a 5-step LangGraph workflow** that enhances the query, retrieves knowledge-base content, keeps only passages an LLM judge scores above 0.8, generates the answer, and routes to a fallback or a scope reply for off-topic questions, with PostgreSQL session storage.
- **Showed the source of every answer** to users and reviewers, down to the document, page, and passage or the web page URL from an allowed domain, and evaluated the chatbot on questions sampled from real users (89.1% Recall@1, 0.93 MRR), with domain experts from customer support and the review team checking each answer and its sources.
- **Logged every workflow step, agent decision, and retrieved source as an audit trail**, with usage tracking.

Senior Project: RAG-Based AI Assistant for Department Website

- **Developed an AI assistant for the department website** that answers questions on courses, faculty, and department information, using FastAPI, Streamlit, LangGraph, and Qdrant with BGE-M3 embeddings over ingested curriculum PDFs and department web pages.
- **Designed a KB-first response workflow** using template-guided generation and Tavily-based web search fallback, with an LLM judge that checked context sufficiency and required a relevance score above 0.9 before a passage was used as a source.

Activities

Global Project-Based Learning

Collaborated with Japanese students from Shibaura Institute of Technology across embedded systems, electrical, and digital logic workshops, communicating in English and Japanese for teamwork and the final presentation.

Technical Skills

Programming & Backend: Python, SQL, JavaScript, TypeScript, PHP, C/C++, HTML/CSS, Node.js, REST APIs, OpenAPI/Swagger

Database & Retrieval Systems: PostgreSQL, MySQL, MongoDB, Database Design, ER Diagrams, Alembic, Qdrant, Chroma, Pinecone, Embedding Models, BGE-M3, Vector Search, Semantic Search, Hybrid Retrieval

AI / ML Engineering: Machine Learning (Regression, Classification, Clustering), Generative AI, Large Language Models (LLMs), LLM APIs (OpenAI, Claude, Gemini), Retrieval-Augmented Generation (RAG), AI Agents, Agentic Workflows, Prompt Engineering, Guardrails, NLP, LLM Evaluation, LLM-as-Judge, Multimodal Document Understanding

Frameworks & Libraries: FastAPI, Flask, Pydantic, pytest, LangChain, LangGraph, Hugging Face, Transformers, pandas, NumPy, scikit-learn, PyTorch, OpenCV, React, Next.js, Streamlit, Gradio, openpyxl

Production & Developer Tools: Docker, Docker Compose, Git, GitHub, GitLab, GitLab CI/CD, AWS S3, MinIO, Postman, Bruno, VS Code, Linux, Jupyter Notebook, Google Colab, Workflow Monitoring

Automation & Data Collection: Web Scraping (Selenium, BeautifulSoup, requests), PDF Parsing (PyMuPDF, pdfplumber, Docling, Marker, Unstructured, MarkItDown), OCR (Tesseract), Search APIs (Tavily, Serper, SerpApi, Exa), Fuzzy Matching, Data Ingestion Pipelines